

Alan Guo

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EDUCATION

University of British Columbia

Vancouver, BC

Engineering Physics, BAsC. | GPA: 4.33/4.33

In Progress

- **Awards:** Trek Excellence Scholarship (2025, 2026); Thomas Beeching Scholarship (2026)
- **Relevant Coursework:** Introduction to Instrument Design, Signals and Systems, Electronic Circuits, Microsensors and Actuators, Experimental Techniques, Software Construction

EXPERIENCE

Agriculture Robot Electrical Design Team Member

Jan. 2026 – Present

UBC Agrobot

Vancouver, BC

- Developed STM32 firmware using PWM to control drive motors and steering actuators for an agricultural robot
- Wired motor drivers, STM32 connections, and electrical components during robot assembly and system integration

Warehouseman

Jun. 2025 – Aug. 2025

Fasteel

Langley, BC

- Gained practical exposure to materials handling, manufacturing workflows, and industrial safety standards

PROJECTS

Autonomous Robotics Design Project

Jun. 2026 – Aug. 2026

- Designed and assembled custom PCBs in KiCad for IR sensing, metal detection, and embedded control, including soldering and hardware bring-up
- Programmed an ESP32 in C++ for motor, sensor, and actuator control, implementing PID tape following and SPI communication with an ADC
- Developed a Python-based vision system on a Raspberry Pi Zero W to detect Teletubby targets and communicate results to the ESP32 over UART
- Integrated power distribution and wiring across microcontrollers, sensors, motor drivers, and actuators, troubleshooting electrical faults with a multimeter and oscilloscope

ESP32 Smart Alarm Clock & Custom PCB

Sep. 2026 – Present

- Designed a custom 2-layer PCB in KiCad integrating an ESP32, OLED display, RTC, ambient-light sensor, buzzer driver, user-input buttons, and USB-C power
- Developed the embedded product from requirements definition and breadboard prototyping through schematic design, PCB layout, assembly, firmware development, and system validation
- Selected electronic components by reviewing datasheets, electrical requirements, and package constraints
- Developed embedded C/C++ firmware for adjustable alarms, time display, menu navigation, snooze functionality, and peripheral control

Servo Motor Circuit

Nov. 2025 – Dec. 2025

- Designed and assembled an operational closed-loop servo control circuit using logic ICs, op-amps, and an R-2R DAC
- Generated LATCH and RESET signals using Schmitt-trigger inverters and RC timing networks
- Implemented and tested an error amplifier to regulate motor speed through feedback control

TECHNICAL SKILLS

Electrical Design: KiCad, Schematic Capture, PCB Layout, PCB Bring-Up, Protoboarding, Soldering, Component Selection, Multimeters, Oscilloscopes

Embedded Systems: ESP32, I²C, SPI, UART, GPIO, PWM, Sensor Integration

Mechanical Design: CAD, Laser Cutting, 3D Printing

Software: C++, C, Java, Python, MATLAB, Excel, Git, Linux